

The status and trends of coronavirus research

A global bibliometric and visualized analysis

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Abstract

Objectives: The infectious pneumonia caused by the Coronavirus Disease 2019 (COVID-19) occurred in Wuhan, Hubei Province, China, from December 2019 and spread the whole country and even other 24 countries. Coronavirus research is of significance to overcome the epidemic. Our study aims to investigate the global status and trends of coronavirus research.

Method: Publications related to the studies of coronavirus research from January 1, 2003 to February 6, 2020 were retrieved from the Science Citation Index-Expanded (SCI-E) of the Web of Science database. A total of 9294 publications were included. The data source was studied and indexed by bibliometric methodology. For visualized study, bibliographic coupling analysis, co-authorship analysis, co-citation analysis, co-occurrence analysis and the analysis of publication trends in coronavirus research were conducted by VOS (visualization of similarities) viewer and GraphPadPrism 6 software.

Results: The number of publications about coronavirus research increased sharply in 2004 for SARS outbreak and increased again in 2012 for MERS outbreak. The USA made the highest contributions to the global research with the most total number of publications, total citation frequency, and the highest H-index, while Netherlands had the highest average citation per item. *Journal of Virology* had the largest publication numbers. The University of Hong Kong is the most contributive institution with the most publications. The main research orientation and funding agency were virology and United States Department of Health Human Services. Keywords of all related studies could be divided into 4 clusters: "Pathological research," "Epidemiology research," "Clinical research," and "Mechanism research."

Conclusions: The outbreak of the epidemic could promote coronavirus research, meanwhile, coronavirus research contributes to overcoming the epidemic. Attention should be drawn to the latest popular research, including "Spike protein," "Receptor binding domain," and "Vaccine." Therefore, more and more efforts will be put into mechanism research and vaccine research and development, which can be helpful to deal with the epidemic.

Abbreviations: COVID-19 = Coronavirus Disease 2019, HCoV-229E = Human Coronavirus 229E, HCoV-HKU1 = Human Coronavirus HKU1, HCoV-NL63 = Human Coronavirus NL63, HCoV-OC43 = Human Coronavirus OC43, IF = impact factor, MERS-CoV = Middle East respiratory syndrome coronavirus, MRI = magnetic resonance imaging, SARS-CoV = severe acute respiratory syndrome coronavirus, SCI-E = Science Citation Index-Expanded, TLS = total link strength, WOS = Web of Science.

Keywords: bibliometrics, coronavirus, MERS, SARS, visualized analysis

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The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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1. Introduction

Coronavirus belongs to order *Nidovirales*, family *Coronaviridae*, and subfamily *Coronaviridae*^[1] with high pathology to mammalian, vertebrates, and avian hosts.^[2] The structure of coronavirus consists of the single-stranded positive-sense viral RNA, Nucleocapsid protein, Envelope protein, Membrane glycoprotein, and Spike protein^[3] (Fig. 1). There have been 7 kinds of coronavirus known to infect humans including Human Coronavirus 229E (HCoV-229E),^[4] Human Coronavirus OC43 (HCoV-OC43),^[5] severe acute respiratory syndrome coronavirus (SARS-CoV),^[6] Human Coronavirus NL63 (HCoV-NL63),^[7] Human Coronavirus HKU1 (HCoV-HKU1),^[8] Middle East respiratory syndrome coronavirus (MERS-CoV),^[9] and Coronavirus Disease 2019 (COVID-19).^[10] SARS-CoV and MERS-CoV, 2 highly transmissible and pathogenic viruses, emerged in humans in 2002 and 2012 respectively and resulted in spread and pathogenic infection in humans.^[11,12] In December 2019, a patient with pneumonia infected by a novel coronavirus was found in Wuhan, China.^[13] Since then, an increasing number of cases of infectious pneumonia have been identified in Wuhan and the whole country.^[14] Now the epidemic has spread abroad.^[15] WHO has named the disease COVID-19, short for "coronavirus disease 2019" in 11 February 2020 (<https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200211-sitrep->

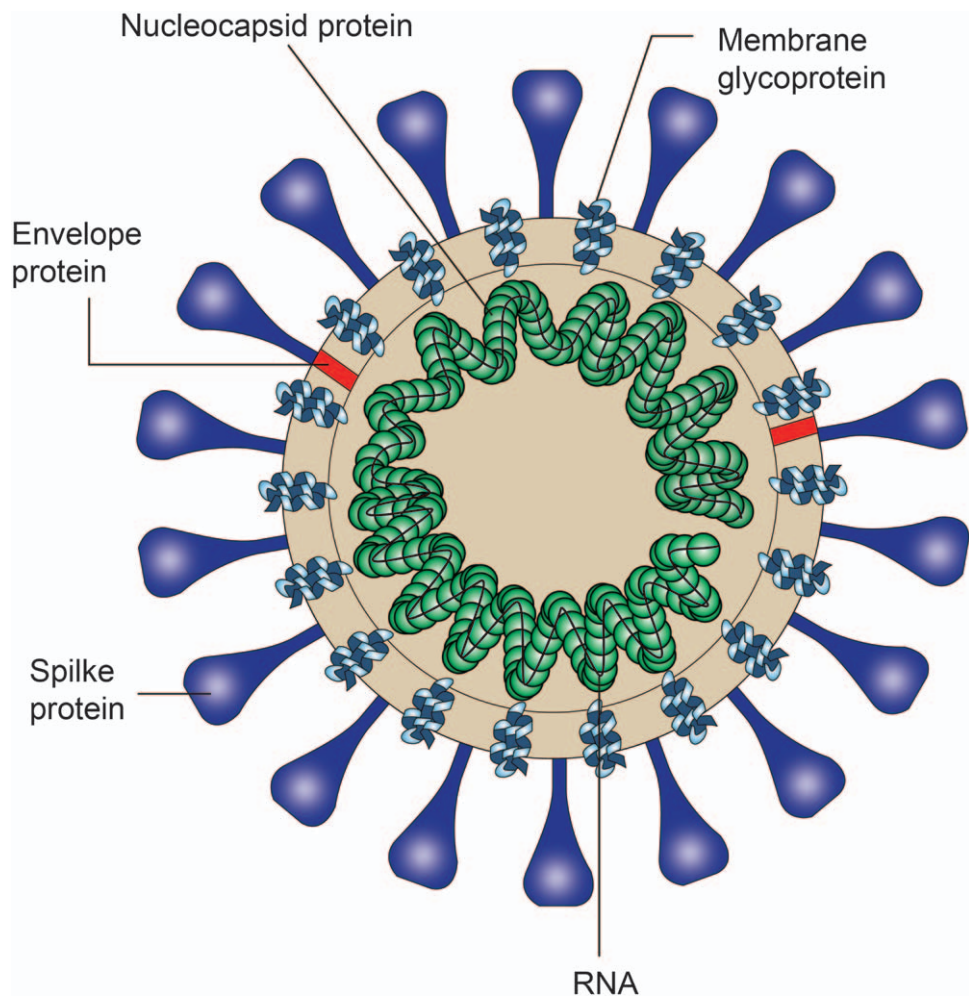


Figure 1. Schematic diagram of the coronavirus structure. The main structure includes the single-stranded positive-sense viral RNA, Nucleocapsid protein, Envelope protein, Membrane glycoprotein, and Spike protein.

22-ncov.pdf?sfvrsn=fb6d49b1_2). China has included the novel coronavirus pneumonia in class B infectious diseases stipulated in the *Law of the People's Republic of China on the Prevention and Treatment of Infectious Diseases* and take preventive and control measures for class A infectious diseases (<http://yuqing.people.com.cn/GB/n1/2020/0121/c209043-31558583.html>). COVID-19 will promote basic and clinical research on coronavirus. However, studies on qualitative and quantitative characteristics of global research of coronavirus are limited. Evaluation of the current status and trends of coronavirus research and predicting promising popular topics and directions in the field are more essential. Publication, as a central part of scientific research, is a significant indicator for research contribution. Bibliometrics combined with visualized mapping has been recognized as an effective means of assessing scientific progress.^[16] Information from online literature databases and metrology characteristics were analyzed via bibliometric analysis, which can be used to quantitatively and qualitatively evaluate the trends in the research community over time.^[17] Comparison among the contributions of scholars, journals, institutions, and countries could be shown through the bibliometric analysis.^[18] Bibliometric analysis is also applied to policy making and clinical guidelines.^[19] Moreover, efficient analysis has been applied

successfully to make studies more intuitional,^[20] including middle East respiratory syndrome coronavirus,^[21] exosomes,^[22] retina regeneration,^[23] stem cells for osteoarthritis,^[24] and spinal ultrasound.^[25] The aim of our study is to assess the status and trends of coronavirus research globally and discover the popular topics for researchers interested in coronavirus.

2. Materials and methods

2.1. Data source

Comprehensive bibliometric data and the SCI-EXPANDED, SSCI, A&HCI, and ESCI citation index database were retrieved from the Web of Science (WOS) Core Collection, which is considered as the optimum database for bibliometrics.^[26] Ethical approval was not required as no human and animal subjects were enrolled.

2.2. Search strategy

The dataset from January 1, 2003 to February 6, 2020 was obtained from the WOS Core Collection. Theme words for searching were referred to MESH terms from PubMed, and the

search term was as follows: (TS=coronavirus) AND (Language=English) AND (Document type=Articles OR Reviews). The information about publications including research orientation, institutions, and funding were ameliorated by the data in the WOS.

2.3. Data collection

The titles, years of publication, names of authors, nationalities, affiliations, keywords, names of publishing journals, abstracts of each record, and citations within the publications downloaded from WOS, were saved as TXT files and imported into Microsoft Excel 2019. Any problem had been solved by consulting experts to reach a consensus.

2.4. Bibliometric and visualized analysis

Bibliometric analysis takes the advantage of bibliometric theory to analyze pertinent literatures through the use of mathematical and statistical approaches, which has become an important tool for global analysis and investigation in various scientific areas.^[27] Basic characteristics of eligible publications, which were mentioned previously, were described through the intrinsic function of WOS. The H-index was regarded as an optimum measurement to evaluate the impact of scientific research and was proposed as an alternative to other bibliometric indicators.^[28] The index of H means that H of a scientist's or a country's publications have been cited in other publications at least H times each,^[29] which reflects both the number of publications and the number of citations per publication.^[30] Impact factors (IFs) of all journals were obtained from the Journal Citation Reports of 2018.

2.5. Analytical methods

GraphPadPrism 6 (GraphPad Software Inc., CA) was used to analyze the data including the time trend of the number of publications, the number of publications from different countries, institutions, funding agencies, research orientations, total citation frequency, average citation frequency, and H-index. VOS viewer (Leiden University, Leiden, Netherlands) is a software tool for the

visualized analysis of the publications,^[31] which was used for bibliographic coupling, co-authorship, co-citation, and co-occurrence analysis.

3. Results

3.1. Analysis of global publications

Variations in the quantity of academic papers on a certain research field are significant indicators for the development trend. Plotting the quantity of publications over time and conducting multivariate statistical analysis contribute to understanding the research level and future trend. A total of 9294 publications from January 1, 2003 to February 6, 2020 were derived from WOS database according to the search criteria. The publication volume of the 18 years has fluctuated, with obvious increase points in 2004 and 2012, which may be related to the corresponding outbreak of SERS and MERS (Fig. 2b). According to publications, the top 20 productive countries were listed in Figure 3a. USA (3225, 34.70%) with the most number was the most contributor and was regarded as the Centers of Disease Control and Prevention, followed by China (2410, 25.93%), Germany (621, 6.68%), and England (558, 6.00%), the top 32 countries that had made the greatest contributions in coronavirus research were shown in Figure 2a, the darker the color, the more the number of publications. As expected, these productive countries are either economically developed or in a rapid development stage, which attach great importance to scientific research. When it comes to the most contributive institutions (Fig. 3b), research orientations (Fig. 3c) and funding (Fig. 3d), the ranking from WOS database were as followed respectively: The University of Hong Kong (452, 4.87%), Chinese Academy of Sciences (323, 3.47%), and Centers for Disease Control and Prevention (197, 2.12%) were listed in the top 3, and Virology (2874, 30.89%), infectious diseases (1217, 13.08%), and Biochemistry molecular biology (1165, 12.51%) were listed in the top 3, and United States Department of Health Human Services (1648, 17.72%), National Institutes of Health, NIH USA (1603, 17.23%), and NIH National Institute of Allergy Infectious Diseases, NIAID (739, 7.94%) were listed in the top 3.

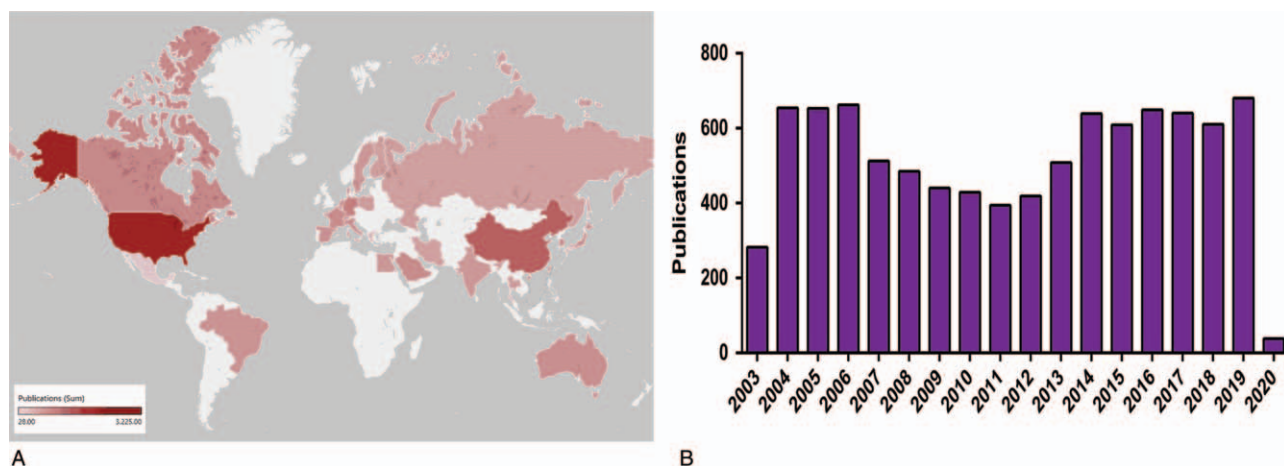


Figure 2. Global trends coronavirus research. (a) World map showing the distribution of coronavirus research, in which the different color depths represent the different numbers of publications in different countries. (b) The annual number of publications related to coronavirus research from January 1, 2003 to February 6, 2020.

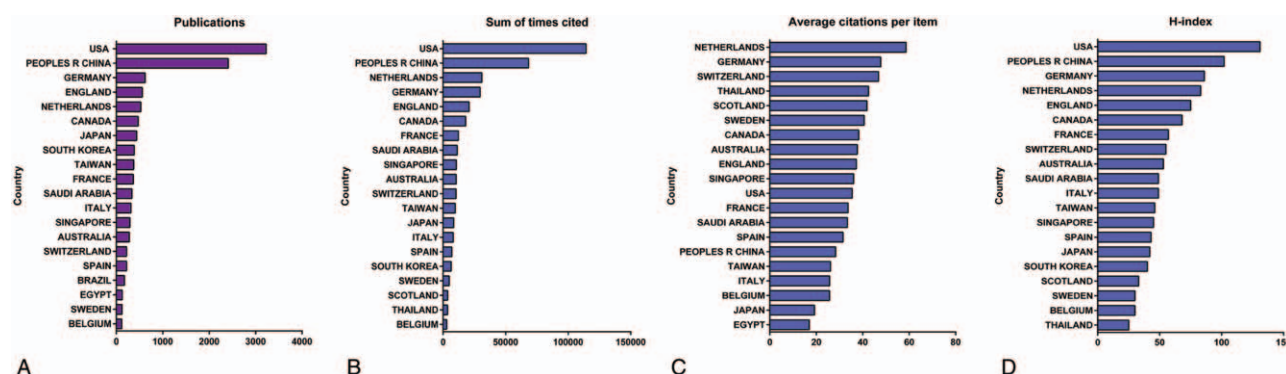


Figure 3. Total number of publications, institutions, research orientations, and funding. (a) The sum of publications related to coronavirus research from 20 countries or regions. (b) The most contributive institution with the most publications in coronavirus research. (c) The most popular orientation about coronavirus research. (d) The major contributive funds for coronavirus research.

3.2. Quality of publications of different countries

In regards to WOS database analysis, we tallied the total citations, average citations, and H-index of each country (Fig. 4).

Papers from USA had the highest number of citations (114,273), followed by China (68,315), Netherlands (30,956). At the same time, Papers from USA had the highest H-index (131). China ranked second in H-index (102), followed by Germany (86), Netherlands (83), and England (75). While the top 5 countries in average citation frequency were Netherlands (58.63), Germany (47.7), Switzerland (46.8), Thailand (43.48), and Scotland (41.81).

3.3. Bibliographic coupling analysis

Bibliographic coupling analysis aims to find the relatedness of items based on the number of references they share and generate the knowledge domain map of main research journals, institutions, and countries in order to demonstrate the collaboration network among them in coronavirus research. In Figure 5, each sphere represents a journal, an institution, and a country, and the sphere size represents the power of total link strength (TLS). The links between sphere represent the associations where

the greater width of the link (namely link strength) means the closer correlation, which is also applied to co-authorship analysis, co-citation analysis, and co-occurrence analysis. Seventy-three journals were shown in Figure 5a in accordance with TLS and the size of sphere representing the power of journal in the research field of coronavirus. The top 5 journals with the greatest total link strength were as follows: *Journal of Virology* (IF=4.324, 2018, TLS=1,410,247 times), *Virology* (IF=2.657, 2018, TLS=538,568 times), *Virus Research* (IF=2.736, 2018, TLS=379,244 times), *Emerging Infectious Diseases* (IF=7.185, 2018, TLS=316,876 times), and *Journal of General Virology* (IF=2.809, 2018, TLS=309,921 times). There were 69 institutions shown in Figure 5b based on papers with the minimum number of documents of an organization more than 50. The University of Hong Kong (TLS=1,747,366 times) ranked the first, followed by Chinese Academy of Sciences (TLS=1,142,965 times), The Chinese University of Hong Kong (TLS=720,073 times), The University of North Carolina (TLS=631,956 times), and National Institute of Allergy and Infectious Diseases (TLS=624,590 times). The top 5 countries based on papers with the minimum number of documents of a country more than 5 were shown in Figure 5c, including USA (TLS=

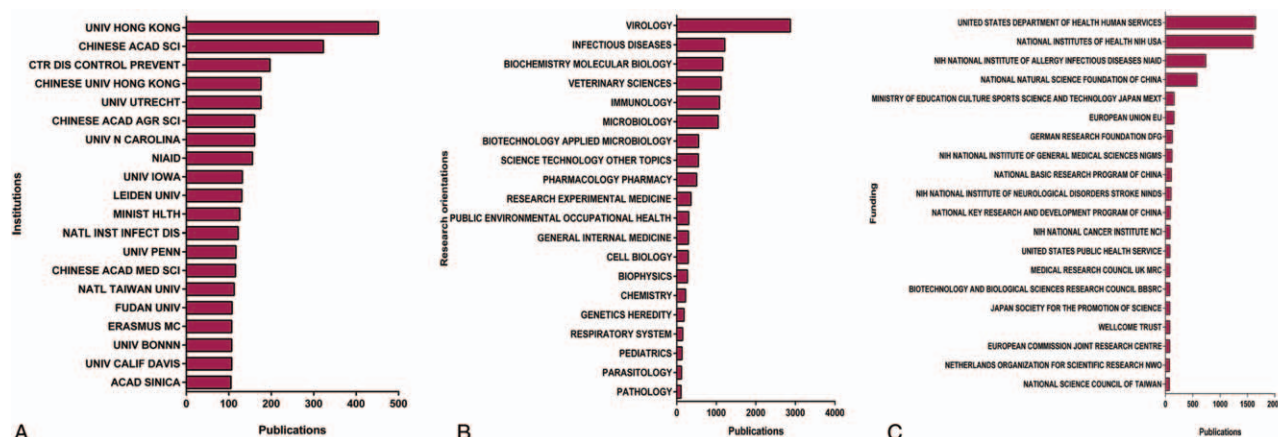


Figure 4. Citation frequency and H-index levels of different countries. (a) The total citations for coronavirus research publications from different countries. (b) The average citations per paper for publications from the different countries. (c) The H-index of publications in the different countries.



Figure 5. Bibliographic coupling analysis of global research about coronavirus. (a) Mapping of the 73 identified journals on coronavirus research. (b) Mapping of the 69 institutions on coronavirus research. (c) Mapping of the 78 countries on coronavirus research. The line between two points in the figure represents that two journals/institutions/countries had established a similarity relationship. The thicker the line, the closer the link between the two journals/institutions/countries.

8,753,964 times), China (TLS=7,863,034 times), Germany (TLS=2,574,147 times), Netherlands (TLS=2,238,094 times), and England (TLS=1,741,562 times).

3.4. Co-authorship analysis

Co-authorship analysis aims to find the relatedness of items based on their number of co-authored documents and generates the knowledge domain map of main research authors, institutions, and countries in order to demonstrate the collaboration network among them in coronavirus research. Creating and analyzing the KDM of co-authorship network of productive authors, institutions, and countries can provide valuable information for individual researchers to seek cooperation partners, for research organizations to develop cooperation groups, and for countries to achieve the goal of academic exchanges. Authors, institutions, and countries were analyzed via VOS viewer and 121 authors, 147 institutions, and 78 countries were shown in Figure 6. Yuen, KY (TLS=598 times), Chan, KH (TLS=411 times), Woo, PCY (TLS=382 times), Lau, SKP (TLS=369 times), and Drosten, C (TLS=243 times) were considered as the top 5 authors with the largest TLS. By the same way, the top 5 institutions with the TLS were as follows: The University of Hong Kong (TLS=341 times), Chinese Academy of Sciences (TLS=288 times), Ministry of Health (TLS=284 times), Alfaisal University (TLS=207 times), and Chinese Academy of Medical Sciences (TLS=203 times). Both authors and institutions were based on papers with the minimum number of documents of an author or organization more than 30. The top 5 countries with largest TLS based on

papers with the minimum number of documents of a country more than 5 were USA (TLS=2045 times), China (TLS=1154 times), Germany (TLS=832 times), England (TLS=762 times), and Netherlands (TLS=653 times).

3.5. Co-citation analysis

Co-citation analysis is purposed to determine the relatedness of items based on the number of times they are cited together. Through co-citation analysis, the important knowledge bases of the research field can be found efficiently and conveniently from the mass of cited references. Furthermore, the relevance of publications can also be analyzed and excavated. Papers with the minimum number of citations of a cited reference more than 100 were analyzed using VOS viewer and 214 references were shown in Figure 7a. The top 5 were as follows: *N. Engl. J. Med.* 2003 May 15;348(20)^[10] (TLS=13,843 times), *N. Engl. J. Med.* 2003 May 15;348(20)^[32] (TLS=13,432 times), *Science.* 2003 May 30;300(5624)^[33] (TLS=11,536 times), *Science.* 2003 May 30;300(5624)^[34] (TLS=10,236 times), *Nature* 2003 Nov 27;426(6965)^[35] (TLS=9232 times). Journals with at least 100 citations were analyzed through VOS viewer and there were 421 journals shown in Figure 7b. The top 5 journals with the largest total link strength were following: *Journal of Virology* (TLS=2,749,603 times), *P. NATL. ACAD. SCI. USA* (TLS=925,219 times), *Virology* (TLS=911,079 times), *Journal of General Virology* (TLS=577,342 times), and *Emerging Infectious Diseases* (TLS=576,144 times). Papers with the minimum number of citations of an author more than 100 were identified



Figure 6. Co-authorship analysis of global research about coronavirus. (a) Mapping of the 121 authors co-authorship analysis on coronavirus research. (b) Mapping of the 147 institutions co-authorship analysis on coronavirus research. (c) Mapping of the 78 countries co-authorship analysis on coronavirus research. The size of the points represents the co-authorship frequency. The line between two points in the figure represents that two authors/institutions/countries had establish collaboration. The thicker the line, the closer the collaboration between the two authors/institutions/countries.



Figure 7. Co-citation analysis of global research about coronavirus. (a) Mapping of co-cited references related to coronavirus research. (b) Mapping of co-cited journals related to coronavirus research. The points with different colors represent the cited references/journals. The size of the points represents the citation frequency. A line between two points means that both were cited in one paper/journal. A shorter line indicates a closer link between two papers/journals. (c) Mapping of the authors co-citation analysis on coronavirus research. Points in the same color belong to the same research direction.

and analyzed via VOS viewer and 467 authors were shown in Figure 7c. The top 5 authors with the greatest total link strength were as follows: Woo, PCY (TLS=57,889 times), Peiris, JSM (TLS=45,852 times), Lau, SKP (TLS=43,016 times), Drosten, C (TLS=39,300 times), and Memish, ZA (TLS=21,369 times).

3.6. Co-occurrence analysis

Co-occurrence analysis purposes to determine the relatedness of items based on the number of documents in which they occur together. Through the analysis, popular topics and directions were identified, which contribute to monitor and follow up the development of science research and programs.^[36,37] Keywords with the minimum number of occurrences more than 80 in all included publications were analyzed via VOS viewer. As shown in Figure 8a, 175 identified keywords were classified into the 4 clusters: “Pathological research,” “Epidemiology research,” “Clinical research,” and “Mechanism research.” In addition to “Coronavirus,” “Virus” and “Infection,” “Identification,” “Acute respiratory syndrome,” “Spike protein,” “SARS,” “Respiratory syndrome coronavirus,” “Replication,” “MERS-coronavirus,” and “Mouse hepatitis virus” were listed in the top with the highest TLS. Keywords were color coded by VOS viewer based on the average time they appeared in the total of 9294 publications (Fig. 8b). The blue color means the keywords appeared early and red colored keywords appeared later. Before 2010, most studies focused on “Epidemiology,” “Diagnosis,”

“Pathogenesis,” “Polymerase chain reaction,” and “Acute respiratory syndrome.” The latest trends showed that “Crystal structure,” “Spike protein,” “Respiratory syndrome coronavirus,” “Receptor binding domain,” and “Vaccine” would be concerned more widely in the future. Papers with the minimum number of occurrences of a keyword more than 80 were identified and analyzed via VOS viewer. Density visualization map (Fig. 8c) was exported by VOS viewer. The times of occurrence of a key word was defined as the color of the area. The larger the times, the warmer the color is. Blue corresponds with the lowest item density and red corresponds with the highest item density.^[31]

4. Discussion

The present study purposes to assess coronavirus research with visualized and bibliometric analysis, which can be applied to present the current status and make predictions in a certain research field. Overall, the number of publications is relatively balanced in the last 18 years, but dramatic progress in coronavirus research especially in 2004 and 2012, which may be due to the outbreak of SARS in 2002 and the outbreak of MERS in 2012. As of February 6, 2020, there have been 38 research reports on coronavirus, COVID-19 will bring coronavirus research to another new climax. At the same time, the results of our study also provide potential popularities in the field, which need further high-quality researches. Moreover, there will



Figure 8. Co-occurrence analysis of global research about coronavirus. (a) Mapping of keywords in the research on coronavirus. The size of the points represents the frequency, and the keywords are divided into 4 clusters: “Pathological research” (upper in green), “Epidemiology research” (right in blue), “Clinical research” (lower in yellow), and “Mechanism research” (left in red). (b) Distribution of keywords according to the mean frequency of appearance. Keywords in blue appeared earlier than those in yellow and red colored keywords appeared later. (c) Density visualization map: the times of occurrence of a key word was defined as the color of the area and the larger the times, the warmer the color is.

be more and more in-depth studies of coronavirus and related articles will be published in the coming years.

Based on the results of the analysis about contributions of countries, USA was the most contributor to publications. In terms of publication number, The University of Hong Kong, Virology, and United States Department of Health Human Services ranked first in institutions, research orientations, and funding agency, respectively. In terms of total number of publications, total citation frequency, and H-index, USA made the highest contributions to the research globally and could be regarded as the pioneer and the leading country in the field of coronavirus research. However, Netherlands had the highest average number of citations and ranked 4th in H-index, Thailand ranked 4th in average number of citations but ranked 20th in H-index, which demonstrated that both Netherlands and Thailand were also excellent contributors in coronavirus research. As a result of the number of related publications, there was a mismatch between H-index and average numbers of citations, both were used for representing the quality of publications and the academic impact of a country. Although China ranked the second only to the USA in these aspects, it was 15th in average citation frequency. The reason why there was a discrepancy between the quantity and quality of publications may be because the Chinese academic evaluation system tends to focus on the quantity of publications instead of quality.^[38] This policy prones researchers to publish articles more quickly, thereby the quality of studies was ignored. With the gradual increase in funding for research in China, the quality of studies will be improved dramatically to keep in step with global publications in the field.

Bibliographic coupling occurs when 2 works contain citations of the third work in their bibliographies. In our study, a similarity relationship among different publications was established by bibliographic coupling analysis in terms of journal, institution, and country. *Journal of Virology*, *Virology*, *Virus Research*, *Emerging Infectious Diseases*, and *Journal of General Virology*, shown in Figure 5a, may be the core journals of coronavirus research. Latest research progress in this area is more likely to be reported by the aforementioned journals. The University of Hong Kong, Chinese Academy of Sciences, and The Chinese University of Hong Kong with the most TLS were regarded as the leader institution in coronavirus research, which also demonstrated that China has made excellent efforts in coronavirus research after SARS. First-class research institutions played a significant and fundamental role in improving the academic level of a country. The authors, shown in Figure 5c, may be the ones who contributed the most in the field and their further studies and latest publications should be closely monitored to obtain the latest advancement in coronavirus research. The collaboration among different countries, institutions, and authors was evaluated by co-authorship analysis and the top of results with higher total link strength showed that the authors/institutions/countries were willing to work with others. For example, Yuen, KY, The University of Hong Kong and USA were the optimum choices for us to cooperate with. Co-citations analysis is utilized to assess the impact of a study by counting the number of times they were cited. The present results indicated that the mechanism studies about coronavirus had the greatest total frequency of citation and many meaningful references were provided. *Journal of Virology* was the journal with the highest citation frequency in the field and the achievement of Woo, PCY, Peiris, JSM and Lau, SKP were widely recognized.

According to the co-occurrence analysis, popular topics and latest research directions were identified. The map of a co-occurrence network was shown in Figure 8a by analyzing the keywords appearing in title and abstract from all included studies. All keywords could be divided into 4 research directions including “Pathological research,” “Epidemiology research,” “Clinical research,” and “Mechanism research.” Our study could make a further clear for the trends of future research, even though the results were consistent with the common sense in the field. The overlay visualization map (Fig. 8b) was the same as the co-occurrence map but in color. Different color stands for different years of the publications. Based on the results, “Crystal structure,” “Spike protein,” “Respiratory syndrome coronavirus,” “Receptor binding domain,” and “Vaccine” may become the next popular subjects in coronavirus research. Meanwhile, the research in the mechanism of related diseases has been emerging widely in recent years. In addition, the clinical research for infectious diseases may become a popular topic in the next few years especially after COVID-19 in Wuhan.

5. Conclusion

The current study presented the global status and trends in coronavirus research. The USA with the most contributions to the research plays a leading role in global research on coronavirus. Although there was a discrepancy among the academic impact, the quality and the quantity of publications from China, China has been making efforts to be an excellent contributor owning outstanding research institutions such as The University of Hong Kong and Chinese Academy of Sciences. Notably, more and more studies about coronavirus will be published in *Journal of Virology*, *Virology*, *Virus Research* in the coming years. The latest research directions, such as “Crystal structure,” “Spike protein,” “Respiratory syndrome coronavirus,” “Receptor binding domain,” and “Vaccine,” are likely to be pioneers in this field and conduct the direction of future studies, which may contribute to recognizing and defeat COVID-19. Much research has been reported on vaccines based on spike proteins for SARS and MERS,^[39,40] which can inspire us to vaccine research for COVID-19 based on spike protein or other structures from COVID-19 and help countermeasures against future spread and pathogenic infection of novel coronavirus in human.

5.1. Strengths and limitations

Publications on coronavirus research in our study were retrieved from the Science Citation Index-Expanded Web of Science database, and the status and trends of coronavirus research were assessed via bibliometric and visualized analysis, which was considered to be relatively objective and comprehensive. However, some limitations in our study have to be mentioned here. Publications in non-English were excluded, resulting in language bias. Additionally, studies published after February 6, 2020 were not included in this study and recently published high-quality articles might not be highlighted as a result of low citation frequency until now, for example *Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus-Infected Pneumonia*. *N. Engl. J. Med.* 2020 Jan 29,^[14] *A new coronavirus associated with human respiratory disease in China*. *Nature*. 2020 Feb 03,^[13] and *A pneumonia outbreak associated with a new coronavirus of probable bat origin*. *Nature*. 2020 Feb 03.^[41] Moreover, the document type was limited to articles and reviews. Therefore,

further research should address the latest studies and other non-English publications, and include other types of documents as complete as possible.

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Author contributions

XM and CX conceived the research; XM and LG collected the data; XM, LG, PF, and CX drafted the article and revised it carefully.

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